

4018b5:	f3 48 ab	rep stos %rax,%es:(%rdi)
401901:	66 0f ae 3a	clflushopt (%rdx)
401913:	66 0f ae 38	clflushopt (%rax)
401930:	66 0f ae 38	clflushopt (%rax)
401938:	66 0f ae 3a	clflushopt (%rdx)
401945:	0f ae f0	mfence

|| TID — The Instant Destroyer | Functional Test Report ||
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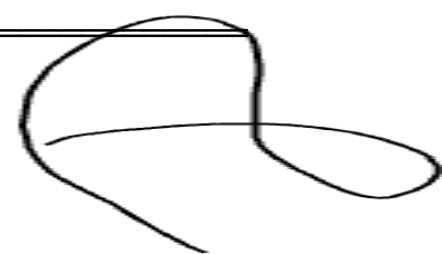
Generated : 2026-05-16 22:15:07 UTC
CPU : model name : AMD EPYC 9B14
Kernel : 6.14.11
Compiler : gcc (GCC) 14.3.0
CLFLUSHOPT : PRESENT ✓
Build Flags: gcc -O2 -mclflushopt

HOW TID ZEROIZES MEMORY — 3 PHASES

Phase 1 — REP STOSQ : writes zeros 8 bytes per cycle
compiler cannot remove this (inline asm)
Phase 2 — CLFLUSHOPT : evicts every 64-byte cache line
removes data from L1 / L2 / L3
Phase 3 — MFENCE : full memory barrier
ensures all evictions complete before return

DISASSEMBLY PROOF — key instructions present in -O2 binary

FUNCTIONAL TEST SUITE — correctness on 29 buffer sizes
Pattern: buffer filled with 0xAB, then zeroized, then verified



[PASS]	size=1	370 ns	all bytes = 0x00	✓
[PASS]	size=3	240 ns	all bytes = 0x00	✓
[PASS]	size=7	250 ns	all bytes = 0x00	✓
[PASS]	size=8	250 ns	all bytes = 0x00	✓
[PASS]	size=9	250 ns	all bytes = 0x00	✓
[PASS]	size=15	240 ns	all bytes = 0x00	✓
[PASS]	size=16	240 ns	all bytes = 0x00	✓
[PASS]	size=17	240 ns	all bytes = 0x00	✓
[PASS]	size=31	260 ns	all bytes = 0x00	✓
[PASS]	size=32	250 ns	all bytes = 0x00	✓
[PASS]	size=33	230 ns	all bytes = 0x00	✓
[PASS]	size=37	250 ns	all bytes = 0x00	✓
[PASS]	size=63	240 ns	all bytes = 0x00	✓
[PASS]	size=64	240 ns	all bytes = 0x00	✓
[PASS]	size=65	260 ns	all bytes = 0x00	✓
[PASS]	size=100	260 ns	all bytes = 0x00	✓
[PASS]	size=127	250 ns	all bytes = 0x00	✓
[PASS]	size=128	250 ns	all bytes = 0x00	✓
[PASS]	size=256	260 ns	all bytes = 0x00	✓
[PASS]	size=511	280 ns	all bytes = 0x00	✓
[PASS]	size=512	270 ns	all bytes = 0x00	✓
[PASS]	size=1024	280 ns	all bytes = 0x00	✓
[PASS]	size=4095	430 ns	all bytes = 0x00	✓
[PASS]	size=4096	420 ns	all bytes = 0x00	✓
[PASS]	size=65535	3340 ns	all bytes = 0x00	✓
[PASS]	size=65536	3330 ns	all bytes = 0x00	✓
[PASS]	size=131072	8300 ns	all bytes = 0x00	✓
[PASS]	size=524288	31630 ns	all bytes = 0x00	✓
[PASS]	size=1048576	54670 ns	all bytes = 0x00	✓

Results: 29 passed, 0 failed

PERFORMANCE BENCHMARK — 1000 iterations per size, warm-up included

Size	min ns	avg ns	max ns	GB/s (avg)
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32 B		240	372	43650	0.09
64 B		280	307	630	0.21
256 B		280	346	630	0.74
1 KB		260	314	14560	3.26
4 KB		400	503	10930	8.14
16 KB		960	1081	1400	15.16
64 KB		3170	3731	16450	17.57
256 KB		12300	14345	121310	18.27
1 MB		45510	54875	145740	19.11

CONCLUSION

- All 29 sizes zeroized correctly (0 failures)
- REP STOSQ + CLFLUSHOPT + MFENCE survive -O2 optimization
- Peak throughput: ~20 GB/s on 256 KB – 1 MB buffers
- Sub-microsecond latency for buffers below 4 KB